A negative answer to a question of Erdős on maximum modulus points

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Abstract

For a non-monomial entire function f, let $\nu_f(r)$ be the number of points on $|z| = r$ at which $|f|$ attains its maximum. Write $f(z) = z^m(c_0 + c_k z^k + O(z^{k+1}))$, where $c_0 c_k \neq 0$ and $k \geq 1$. We prove that $\nu_f(r) \leq 2k$ outside a countable set of radii. In particular, $\nu_f(r)$ cannot tend to infinity, answering the second question of Erdős on maximum modulus points. The proof studies the correspondence $A(z) = \overline{A(\overline{w})}$, where $A = zf'/f$, using implicit-function continuation, growth in a direct tract, and a finite proper map of analytic curves.

1 Introduction

For a nonzero entire function f, put

$$M(r, f) = \max_{|z|=r} |f(z)|, \qquad \nu_f(r) = \#\{z : |z| = r,\ |f(z)| = M(r, f)\}.$$

If f is not a monomial, then $\nu_f(r)$ is finite for every $r > 0$.[1] Erdős asked whether a single such function can satisfy

$$\limsup_{r\to\infty} \nu_f(r) = \infty, \qquad \text{and whether it can satisfy} \qquad \liminf_{r\to\infty} \nu_f(r) = \infty;$$

see [3, 6] for the problem and its history. Herzog and Piranian [8] answered the first question affirmatively; see also the account in [6]. Pardo-Simón and Sixsmith [9] obtained such an example of finite order. Glücksam and Pardo-Simón [6] constructed an approximate analogue of the second property, with many separated arcs on which the modulus is close to its maximum.

There is also a classical local theory near the origin. Dividing f in (1.1) by $c_0 z^m$ leaves its maximum modulus points unchanged and gives the normalization $1 + az^k + O(z^{k+1})$. Hayman [7, Theorem I(iii), pp. 142–143] proved that its maximum modulus set near zero consists of at most k analytic curves. Evdoridou, Pardo-Simón, and Sixsmith [5, Theorem 1.4(b)] show that each curve meets every sufficiently small circle exactly once. It follows that $\nu_f(r) \leq k$ for all sufficiently small $r > 0$. Theorem 1.1 is a global counterpart of this local bound: the same k controls the count at all but countably many radii. The unbounded upper limits in [8, 9] occur on the exceptional radii.

We prove that the answer to the second question is negative.

[1] If $|f| = c > 0$ on an arc of $|z| = r$, the identity theorem gives $f(z)f^{\#}(r^2/z) = c^2$ on $\mathbb{C}^*$, where $f^{\#}(z) = \overline{f(\overline{z})}$. Writing $f = z^m h$, $h(0) \neq 0$, and letting $z \to \infty$ gives $f(z) = O(|z|^m)$. Thus f is a polynomial of degree at most m, hence a monomial. For non-monomial f, the real-analytic function $\theta \mapsto |f(re^{i\theta})|^2$ has a discrete maximum set on the compact circle, hence finitely many maximum points.

Theorem 1.1. *Let f be a nonzero entire function that is not a monomial, and write*

$$f(z) = z^m(c_0 + c_k z^k + O(z^{k+1})), \qquad m \geq 0, \quad k \geq 1, \quad c_0 c_k \neq 0. \tag{1.1}$$

There is a countable set $E \subset (0, \infty)$ such that

$$\nu_f(r) \leq 2k \qquad (r \notin E). \tag{1.2}$$

Consequently,

$$\liminf_{r \to \infty} \nu_f(r) \leq 2k < \infty.$$

The integer k is the difference between the degrees of the first two nonzero terms of f. The factor 2 is attained by an explicit example given in Remark 5.1.

Set $A = zf'/f$. At radii where $\log M(r, f)$ is differentiable with respect to $\log r$, all maximum modulus points have the same real value of A. Hence they determine points $(z, \overline{z})$ on the correspondence $A(z) = \overline{A(\overline{w})}$ with the same values of zw and $A(z)$. We show that every irreducible component of this correspondence contains points with $|zw|$ arbitrarily small. Since $A - m$ has local degree k at the origin, the generic fibres of the resulting finite proper map have size at most $2k$.

2 Analytic preliminaries

Lemma 2.1 (Stoïlow [10]; see also [4])**.** *Let F be entire on $\mathbb{C}^2$, and suppose that*

$$F(a, b) = 0, \qquad F_w(a, b) \neq 0.$$

The projection to the z-plane of the maximal analytic continuation of the implicit germ $w = w(z)$ through (a, b) is dense in $\mathbb{C}$.

Following Bergweiler, Rippon, and Stallard [1, Definition 2.1], an unbounded domain $D \subset \mathbb{C}$, with unbounded complement and piecewise smooth boundary, is a *direct tract* of Q if Q is holomorphic in D, continuous on its closure, and

$$|Q| = a \quad \text{on } \partial D, \qquad |Q| > a \quad \text{in } D$$

for some $a > 0$.

Lemma 2.2 (Fuchs; see [1, Theorem 2.1])**.** *Let D be a direct tract of Q, with boundary value a, and put*

$$u(w) = \begin{cases} \log(|Q(w)|/a), & w \in D, \\ 0, & w \notin D. \end{cases}$$

Then u is a continuous subharmonic function on $\mathbb{C}$, and

$$\frac{\max_{|w|=R} u(w)}{\log R} \longrightarrow \infty \qquad (R \to \infty). \tag{2.1}$$

3 Small products on the correspondence

Fix f, m, k as in Theorem 1.1. Write $f = z^m h$, where $h(0) \neq 0$, and define the meromorphic functions

$$A(z) = \frac{zf'(z)}{f(z)} = m + \frac{zh'(z)}{h(z)}, \qquad B(w) = A^{\#}(w) := \overline{A(\overline{w})}. \tag{3.1}$$

The removable singularity of A at zero is filled in. The function A is nonconstant, since otherwise f would be a monomial. Moreover, (1.1) gives

$$A(z) - m = \frac{kc_k}{c_0} z^k + O(z^{k+1}).$$

In particular, $k = \operatorname{ord}_0(A - m)$.

There is a holomorphic local coordinate χ at zero such that

$$\chi(0) = 0, \qquad \chi'(0) \neq 0, \qquad A(z) - m = \chi(z)^k. \tag{3.2}$$

Choose $\delta \in (0, 1)$ small enough that

$$V = \chi^{-1}\{\xi : |\xi| < \delta^{1/k}\}$$

has compact closure in the coordinate neighbourhood. For every $|\lambda - m| < \delta$, the equation $A(z) = \lambda$ has exactly k roots in V, counted with multiplicity. They are distinct when $\lambda \neq m$, and there is a constant C such that all of them satisfy

$$|z| \leq C|\lambda - m|^{1/k}. \tag{3.3}$$

The corresponding statements hold for B on $V^{\#} = \{\overline{z} : z \in V\}$.

Choose δ also so that it is not the modulus of any finite, nonzero critical value of $B - m$. Then $\{|B - m| = \delta\}$ is a locally finite union of smooth curves.

Choose entire functions p, q with no common zero such that $A = p/q$ and $q(0) \neq 0$. Set

$$F(z, w) = p(z)q^{\#}(w) - p^{\#}(w)q(z), \qquad \mathcal{X} = \{(z, w) \in \mathbb{C}^2 : F(z, w) = 0\}. \tag{3.4}$$

All analytic sets in this paper are given their reduced structure. Away from poles, the equation in (3.4) is $A(z) = B(w)$. Since p, q have no common zero and A is nonconstant, $\mathcal{X}$ has no horizontal or vertical irreducible component. At points with finite common value,

$$F_w(z, w) = -q(z)q^{\#}(w)B'(w).$$

It follows that $F_w \neq 0$ at a general point of each irreducible component: the coordinate w is nonconstant there, whereas the zeros of B' are discrete.

Lemma 3.1. *Every irreducible component X of $\mathcal{X}$ contains a sequence (z_j, w_j) such that*

$$z_j w_j \neq 0, \qquad A(z_j) = B(w_j) \in \mathbb{C} \setminus \{m\}, \qquad |z_j w_j| \longrightarrow 0. \tag{3.5}$$

Proof. Choose a regular point of X at which $F_w \neq 0$ and apply Lemma 2.1 to the corresponding implicit germ. The continued graph stays in X by the identity theorem on its connected Riemann surface of analytic continuation. Since its projection is dense, the continuation reaches a point $(z_0, w_0) \in X$ with

$$z_0 \in V \setminus \{0\}, \qquad 0 < |A(z_0) - m| < \delta.$$

The value $B(w_0) = A(z_0)$ is finite: since $q(z_0) \neq 0$, $F(z_0, w_0) = 0$ and the absence of common zeros of $p^{\#}, q^{\#}$ exclude a pole at w_0. Let U be the connected component of $\{|B - m| < \delta\}$ containing w_0. In $V \times U$, consider the analytic curve

$$\chi(z)^k = B(w) - m. \tag{3.6}$$

Its projection to U is finite and proper. Indeed, over a compact subset of U, the right-hand side remains in a compact subset of $\{\xi : |\xi| < \delta\}$, so all its k-th roots remain in a compact subset of the coordinate disc.

Since $A - m = \chi^k$, with $\chi(z_0) \neq 0$ and $\chi'(z_0) \neq 0$, we have $A'(z_0) \neq 0$. Thus the curve (3.6) is smooth at (z_0, w_0). Let Y be its irreducible component through this point. Near (z_0, w_0), it is the same branch as X, hence $Y \subset X$ by the identity theorem for analytic sets. The projection $Y \to U$ is finite, proper, and nonconstant. Its image is therefore both open and closed in U, so the projection is surjective. Thus every $w \in U$ has a lift $(z, w) \in Y$, with z satisfying (3.3).

If $B(b) = m$ for some $b \in U$, choose $w_j \to b$ in U, with $w_j \neq 0$ and $B(w_j) \neq m$, and lift these points to Y. Equation (3.3) gives $z_j \to 0$, and hence $z_j w_j \to 0$. This proves the lemma in this case.

Suppose now that $B - m$ has no zero in U, and put

$$Q(w) = \frac{\delta}{B(w) - m}. \tag{3.7}$$

Every finite boundary point of U satisfies $|B - m| = \delta$. Thus Q is holomorphic in U, continuous on $\overline{U}$, and satisfies

$$|Q| > 1 \quad \text{in } U, \qquad |Q| = 1 \quad \text{on } \partial U.$$

The maximum principle shows that U is unbounded.

We claim that $\mathbb{C} \setminus U$ is unbounded. Otherwise U contains a neighbourhood of infinity, on which $|B - m| < \delta < 1$. Thus B is meromorphic on $\mathbb{C}$ and holomorphic and bounded near infinity, hence rational; so is $A = B^{\#}$. Since $f'/f = A/z$ is rational, factoring out the finitely many zeros of f gives $f = Pe^R$ for polynomials P, R. Therefore $A = zP'/P + zR'$. Boundedness at infinity forces R to be constant, so $A \to \deg P$; the bound $|A - m| < \delta < 1$ gives $\deg P = m$. Since z^m divides P, f is a monomial, a contradiction.

Consequently, U is a direct tract of Q with boundary value one. Define $u = \log |Q|$ in U, and $u = 0$ outside U. For each sufficiently large R, choose w_R on $|w| = R$ at which u attains its maximum. By Lemma 2.2, this maximum is positive and

$$\frac{u(w_R)}{\log R} \longrightarrow \infty.$$

In particular, $w_R \in U$. Choose a lift $(z_R, w_R) \in Y$. Equations (3.3) and (3.7) yield

$$|z_R w_R| \leq C\delta^{1/k} R \exp\left(-\frac{u(w_R)}{k}\right) \longrightarrow 0. \tag{3.8}$$

Since $B - m$ has no zero in U, these lifts have nonzero coordinates and common value different from m. They satisfy (3.5). □

4 A finite map of analytic curves

Remove the zero coordinates, poles, and common value m, and set

$$\begin{aligned} \mathcal{X}^* &= \{(z, w) \in \mathcal{X} : zw \neq 0,\ A(z) = B(w) \in \mathbb{C} \setminus \{m\}\}, \\ \mathcal{B} &= \mathbb{C}^* \times (\mathbb{C} \setminus \{m\}). \end{aligned}$$

On each irreducible component of $\mathcal{X}$, the deleted locus is discrete: the absence of horizontal or vertical components ensures that none is contained in $zw = 0$, $A(z) = m$, or the pole loci. Thus every irreducible component of $\mathcal{X}^*$ is obtained from one of $\mathcal{X}$ by deleting a discrete set. Consider the holomorphic map

$$\Phi : \mathcal{X}^* \longrightarrow \mathcal{B}, \qquad \Phi(z,w) = (zw, A(z)) = (s,\lambda). \tag{4.1}$$

Lemma 4.1. *The map Φ is proper and has finite fibres.*

Proof. Suppose that (s,λ) remains in a compact subset of $\mathcal{B}$. Then $|s|$ is bounded above and away from zero, while λ is bounded and stays a positive distance from m. If a sequence of preimages had $z \to \infty$, then $w = s/z \to 0$, and therefore $\lambda = B(w) \to m$, a contradiction. If $z \to 0$, then $\lambda = A(z) \to m$, again a contradiction. The same arguments apply with z and w interchanged. Neither coordinate can approach a pole, since λ is bounded. It follows that the inverse image of the compact set is compact in $\mathcal{X}^*$.

For fixed (s,λ), the possible z-coordinates form a compact subset of the discrete zero set of $A - \lambda$. There are therefore only finitely many of them, and each determines $w = s/z$ uniquely. $\square$

Proposition 4.2. *The image $\Gamma = \Phi(\mathcal{X}^*)$ is a closed analytic curve in $\mathcal{B}$. There is a locally finite set $T \subset \Gamma$ such that*

$$\#\Phi^{-1}(s,\lambda) \le 2k \qquad ((s,\lambda) \in \Gamma \setminus T). \tag{4.2}$$

Proof. By Remmert's proper mapping theorem [2, Chapter II, Theorem 8.8, p. 118] and Lemma 4.1, Γ is a closed analytic curve. Let T consist of the singular points of Γ together with the images of the singular and ramification points of $\mathcal{X}^*$, where ramification is understood on the normalization of each source component [2, Chapter II, Theorem 7.12, pp. 114–115]. The map is nonconstant on every source component, so its ramification locus is discrete; properness then makes these exceptional sets locally finite in Γ.

For each irreducible component Γ_i of Γ, the map

$$\Phi^{-1}(\Gamma_i \setminus T) \longrightarrow \Gamma_i \setminus T$$

is proper and, off T, a local biholomorphism, hence a finite covering. Points of $\Gamma_i \setminus T$ lie on no other component of Γ, since intersections of components are singular. The normalization of Γ_i is a connected Riemann surface, and deleting a locally finite set leaves it connected. Since T contains the singular points of Γ, it follows that $\Gamma_i \setminus T$ is connected. Hence there is an integer d_i such that

$$\#\Phi^{-1}(s,\lambda) = d_i \qquad ((s,\lambda) \in \Gamma_i \setminus T). \tag{4.3}$$

Choose $\eta > 0$ so that the disc $\{|z| < \eta\}$ is contained in both V and $V^{\#}$. If $0 < |s| < \eta^2$, every preimage satisfies

$$\min\{|z|, |w|\} < \eta.$$

There are at most k preimages with $|z| < \eta$: the existence of such a coordinate implies $|\lambda - m| < \delta$, and the local root count in Section 3 applies. Each root determines the other coordinate from $w = s/z$. Likewise there are at most k preimages with $|w| < \eta$. Hence

$$\#\Phi^{-1}(s,\lambda) \le 2k \qquad (0 < |s| < \eta^2). \tag{4.4}$$

The image of each irreducible component of $\mathcal{X}^*$ is an irreducible component of Γ, and every component of Γ arises in this way. Lemma 3.1 therefore shows that each Γ_i meets $\{0 < |s| < \eta^2\}$. This intersection is a nonempty open subset of a curve, so it is not contained in T. Evaluating (4.3) at a point of this intersection and using (4.4) gives $d_i \le 2k$. Equation (4.2) follows for every component. $\square$

5 Maximum modulus points

Proof of Theorem 1.1. Set

$$v(x) = \log M(e^x, f), \qquad x \in \mathbb{R}.$$

Hadamard's three-circles theorem makes v convex. Its set N of nondifferentiability points is therefore at most countable. Suppose that $x \notin N$, let $r = e^x$, and let $z = re^{i\theta}$ be a maximum modulus point. Since $f(z) \neq 0$, angular differentiation gives

$$0 = \frac{\partial}{\partial \theta} \log |f(re^{i\theta})| = -\operatorname{Im} A(z). \tag{5.1}$$

For this fixed θ, the function $t \mapsto \log |f(e^{t+i\theta})|$ lies below $v(t)$ and agrees with it at $t = x$. Differentiating at this contact point yields

$$\operatorname{Re} A(z) = v'(x). \tag{5.2}$$

Thus all maximum modulus points on the circle have the same real value $A(z) = \lambda := v'(x)$.

In fact $\lambda > m$. To see this, write

$$b(x) = v(x) - mx = \log M(e^x, h).$$

The function b is convex and strictly increasing, since the maximum modulus of a nonconstant entire function strictly increases with radius. For any $t < x$, the convex secant-slope inequality gives

$$0 < \frac{b(x) - b(t)}{x - t} \leq b'(x).$$

Hence $\lambda = m + b'(x) > m$.

For every maximum modulus point z on $|z| = r$, reality of λ gives $B(\overline{z}) = \overline{A(z)} = \lambda$. Since $\lambda \neq m$ and $r^2 \neq 0$, we obtain

$$(z, \overline{z}) \in \mathcal{X}^*, \qquad \Phi(z, \overline{z}) = (r^2, \lambda). \tag{5.3}$$

Distinct maximum points give distinct points of this fibre. If $(r^2, \lambda) \notin T$, Proposition 4.2 gives $\nu_f(r) \leq 2k$.

Since T is locally finite, it is countable. Define

$$E = \{e^x : x \in N\} \ \cup \ \{\sqrt{s} : s > 0 \text{ and } (s, \lambda) \in T \text{ for some } \lambda\}.$$

This is a countable set of positive radii, and (1.2) holds for every $r \notin E$. Since every interval (R, ∞) contains radii outside E, the asserted bound on the lower limit follows. □

Remark 5.1. For $k \geq 1$, the function $f(z) = \exp(2z^k - z^{2k})$ has first two nonzero Taylor terms $1 + 2z^k$. Put $R = r^k$ and $\phi = k\theta$. Then

$$\log |f(re^{i\theta})| = R^2 + \frac{1}{2} - 2R^2 \left(\cos \phi - \frac{1}{2R} \right)^2 .$$

For $0 < R \leq 1/2$, the maximum is attained when $\cos(k\theta) = 1$, giving k distinct points. For $R > 1/2$, it is attained precisely when $\cos(k\theta) = 1/(2R)$, giving $2k$ distinct points. Thus the bound in Theorem 1.1 is sharp for every k.

Use of generative AI

The proofs were generated with GPT-6 (Codex) and checked step by step by the author. The arguments of Section 5, the local fibre count (4.4), and Remark 5.1 are formalized in Lean 4 with Mathlib at `https://github.com/FireflySentinel/erdos-1117`. Lemma 3.1, Lemma 4.1, and Proposition 4.2 are not formalized; their global fibre estimate enters the final Lean theorem as the hypothesis `GlobalFibreBound`.

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